MUNICIPAL CORPORATION OF GREATER MUMBAI (MCGM / BMC)

ORANGE ALERT

TIER_2_SCREENING_SIMULATION

DISASTER MANAGEMENT CELL • EMERGENCY OPERATIONS EXECUTIVE DOSSIER

Document ID: **DOS-COASTAL_MONSOON_SURGE-T6H-MVGFHGQO** • Generated: Tuesday, 20 October 2026 at 2:05 pm IST

EXECUTIVE SITUATION SUMMARY:

Multi-criteria hydrodynamic simulation evaluated for scenario **"Scenario C: Arabian Sea High Tide & Storm Sluice Interlock"** across all 24 canonical administrative wards of Greater Mumbai over a **T+6h** forecast horizon. Topographical saucer basins and coastal backpressure were evaluated against Copernicus GLO-30 DEM terrain profiles.

SEVERE WARDS	HIGH RISK WARDS	CITYWIDE MEAN RISK	SCREENED POPULATION
0	20	48.4	~1004k
Score ≥ 80 / 100	Score 60 - 79 / 100	Normalised 0-100 scale	Low-lying contour buffers

1. KEY INTELLIGENCE FINDINGS

EVIDENCE-BASED SYNTHESIS

- Peak hazard susceptibility is concentrated in Ward A (Colaba / Fort / Churchgate) at a score of 50/100.
- Top intervention priorities across Mumbai's 24 administrative wards: Ward A (50/100), Ward B (50/100), Ward C (50/100), Ward D (50/100), Ward E (50/100).
- Estimated population residing within modeled low-lying screening zones is ~10,03,600 across all wards.
- Critical infrastructure in proximity includes 6 tertiary hospitals and 11 major arterial transit segments.
- Overall municipal alert level is classified as ORANGE_ALERT under applied parameters.

2. HIGHEST PRIORITY INTERVENTION WARDS

TOP 5 OF 24

RANK	WARD ID	WARD DESIGNATION	RISK SCORE	SEVERITY TIER	PRIMARY PHYSICAL TRIGGER
#1	Ward A	Ward A (Colaba / Fort / Churchgate)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#2	Ward B	Ward B (Sandhurst Road / Dongri / Bhendi Bazar)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#3	Ward C	Ward C (Marine Lines / Kalbadevi / Bhuleshwar)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#4	Ward D	Ward D (Grant Road / Malabar Hill / Walkeshwar)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#5	Ward E	Ward E (Byculla / Nagpada / Mazgaon)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater

3. POTENTIALLY EXPOSED MUNICIPAL INFRASTRUCTURE

CRITICAL ASSETS

Tertiary Municipal Hospitals — 6 assets

KEM Hospital, Sion Hospital, Nair Hospital, Cooper Hospital

Caveat: Located in spatial screening proximity (<500m); physical water ingress not confirmed.

Arterial Transit Corridors — 11 assets

Dr. Babasaheb Ambedkar Road, LBS Marg, Swami Vivekananda (SV) Road

Caveat: Road centerline intersects modeled basin buffer; traffic impassability subject to spot depth.

Copernicus GLO-30 Topographic Depressions **1 assets**

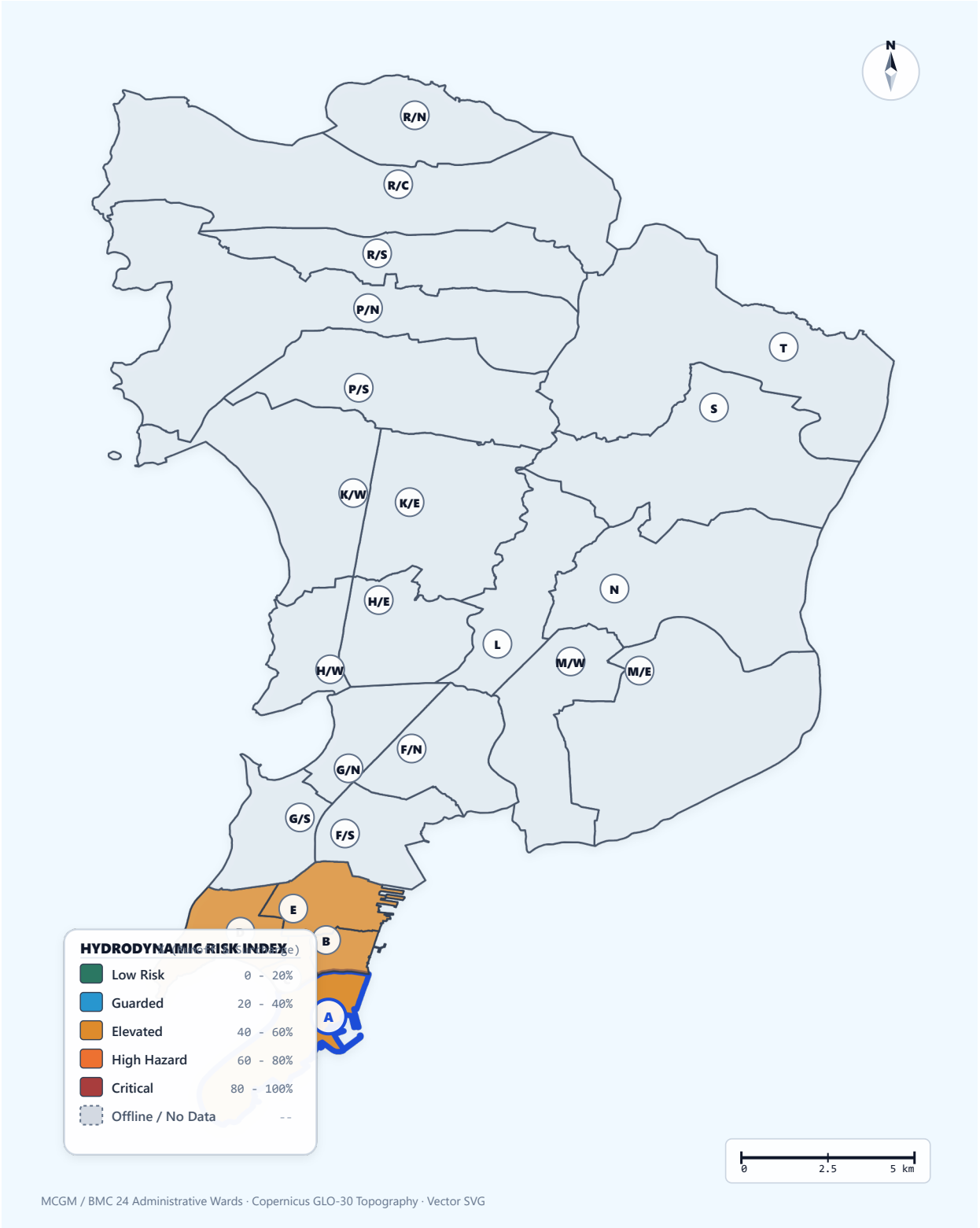
Hindmata Basin (4.5m ASL), Kurla Sump (5.2m ASL), Milan Subway (5.8m ASL)

Caveat: Documented elevation minima where gravity storm-water discharge reverses during high tide.

GREATER MUMBAI 24-WARD HYDRODYNAMIC CARTOGRAPHY

T+6h Simulation Horizon

FIGURE 1.1: MUNICIPAL CHOROPLETH RISK DISTRIBUTION & TOPOGRAPHIC SINK SCREENING



Cartographic Note: Wards are filled according to their calculated waterlogging and drainage surcharge score for the active scenario. Boundaries adhere to canonical BMC administrative boundaries (EPSG:4326). Blue boundary highlights selected ward focus. Shaded contours represent Copernicus GLO-30 DEM depression sinks.

COMPREHENSIVE 24-WARD MUNICIPAL EVALUATION MATRIX

24 BMC Administrative Wards

SECTION C: ADMINISTRATIVE WARD RANKINGS & POPULATION EXPOSURE

RANK	WARD	ADMINISTRATIVE WARD DESIGNATION	SCORE	SEVERITY	ELEV (M)	EXPOSED POP	ASSETS	PRIMARY TRIGGER
#1	A	Ward A (Colaba / Fort / Churchgate)	50	HIGH SINKING / CHOKE	5.2m	~14,800	7	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.15m
#2	B	Ward B (Sandhurst Road / Dongri / Bhendi Bazar)	50	HIGH SINKING / CHOKE	6.8m	~11,200	2	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.15m
#3	C	Ward C (Marine Lines / Kalbadevi / Bhuleshwar)	50	HIGH SINKING / CHOKE	5.8m	~15,200	0	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.15m
#4	D	Ward D (Grant Road / Malabar Hill / Walkeshwar)	50	HIGH SINKING / CHOKE	14.5m	~30,400	4	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.15m
#5	E	Ward E (Byculla / Nagpada / Mazgaon)	50	HIGH SINKING / CHOKE	6.2m	~35,200	4	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.15m

* Population estimates derive from 2011 Census ward totals adjusted to 2026 BMC projections, intersected with low-lying saucer depression buffer zones.

WARD FOCUS: WARD A (COLABA / FORT / CHURCHGATE)

HIGH SINKING / CHOKE (50 / 100)

SECTION D: TOPOGRAPHIC PROFILE & CRITICAL ASSET EXPOSURE

ADMINISTRATIVE ZONE	MEAN GROUND ELEVATION	DEPRESSION HOTSPOTS	EXPOSED POPULATION
Zone I (South Mumbai / Island City South) Coastal Backwater Zone	5.2m Copernicus GLO-30 ASL	0 Chronic low points	~14,800 of ~1,85,000 residents

TOPOGRAPHICAL SINKS & DRAINAGE OUTFALLS

Drainage Waterways & Basins: Arabian Sea South & Mumbai Harbour Outfalls, Arabian Sea Coastline

Depression Sinks: None identified

Historical Chronic Waterlogging Points: Colaba Post Office junction, Ballard Estate / GPO Subway, Regal Cinema Junction

POTENTIALLY EXPOSED MEDICAL & TRANSPORT INFRASTRUCTURE

HOSPITALS IN LOW-LYING BUFFER

- **Public Health Department - Government of Maharashtra (State Secretariat)** (State Apex Health Directorate & Policy Secretariat)
- **Directorate of Health Services (DHS Maharashtra) & NHM Headquarters** (State Health Services Directorate & NHM Command Center)
- **Cama & Albless Hospital for Women and Children** (State Specialized Maternal & Pediatric Hospital)

MAJOR ARTERIALS IN INUNDATION CORRIDOR

- **Marine Drive (Netaji Subhash Chandra Bose Road)** (primary)
- **Shahid Bhagat Singh Road / Colaba Causeway** (primary)
- **Veer Nariman Road (Churchgate to Flora Fountain)** (secondary)
- **Mahatma Gandhi Road (Kala Ghoda to Metro Cinema)** (primary)

METHODOLOGY & EXPOSURE CAVEAT:

Exposure figures represent geographic co-location within digital elevation sink buffers and BRIMSTOWAD catchments. They do not constitute verified flooding or structural damage. Actual inundation is subject to local stormwater drain maintenance, operational dewatering pumps, and real-time gate configurations.

MATHEMATICAL METHODOLOGY & INPUT CONDITIONS

SECTION E: FORMULA SPECIFICATION, TERRAIN INTEGRATION & AUDIT TRACE

1. MATHEMATICAL FORMULATION

HYDRODYNAMIC SCREENING

```
If HighTide >= 4.5m, SluiceState = CLOSED; ResidualDischarge = min(PumpingCapacity, Inflow); CoastalImpact = S_base * (1 + 0.65 * (HighTide - 4.5) / 1.0)
```

Latex Representation: `If HighTide >= 4.5m, SluiceState = CLOSED; ResidualDischarge = min(PumpingCapacity, Inflow); CoastalImpact = S_base * (1 + 0.65 * (HighTide - 4.5) / 1.0)`
Baseline Definition: Dry-weather baseline assuming 0 mm/h rain, 1.5m astronomical tide, 0% drainage blockage, and historical hotspot background. Delta percent represents: ((Score - Baseline) / max(1, Baseline)) * 100%.
Spatial Resolution: 24 Canonical BMC Administrative Wards + 250m Continuous Raster Grid • **Temporal Resolution:** Hourly increments from T+0h to T+6h

2. USER-CONTROLLED SCENARIO INPUTS & OBSERVATIONS

PARAMETER AUDIT

INPUT PARAMETER / VARIABLE	VALUE	UNIT	STATUS	SOURCE / NOTES
Rainfall Intensity	50	mm/h	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Astronomical High Tide Height	4.65	meters	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Arabian Sea Swell / Surge	2.4	meters	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Automatic Weather Station (AWS) Precipitation (Live Telemetry)	0	mm/h	ACTIVE	Open-Meteo High-Resolution AWS Feed
Apollo Bandar Astronomical Tide Level (Live Telemetry)	1.85	meters Chart Datum (CD)	ACTIVE	Mumbai Port Trust (MbPT) Tide Gauge Table

3. MISSING OR OFFLINE TELEMETRY FEEDS

HONEST DISCLOSURE

VARIABLE NAME	IMPACT ON SIMULATION MODEL	DETERMINISTIC FALLBACK USED
Real-time Drainage Sluice Gate SCADA	Outfall gate blockage % is user-configured rather than live sensor-driven.	Scenario parameter default applied with explicit assumption note.
Mithi River Ultrasonic Level Sensors	River corridor ponding modeled via static Copernicus elevation buffer.	GLO-30 DEM 5.0m elevation contour threshold.

RECOMMENDED ACTION DIRECTIVES & STATUTORY DISCLAIMER

SECTION F: INTER-AGENCY OPERATIONS & EXECUTIVE SIGN-OFF

1. DEPARTMENTAL OPERATIONAL ACTION DIRECTIVES

PRIORITY INTERVENTIONS

PHASE	DEPT	ACTION TITLE & DIRECTIVES	TARGET WARDS	PRIORITY
BEFORE_EVENT	SWD	**Pre-position mobile pumping units at high-risk sluices** Deploy additional mobile pump units to Ballard Bunder, Mazgaon Outfall, Haji Ali, and Britannia stations and stage them within 2 km of each facility to augment capacity if surge exceeds design thresholds.	A, B, D, E	CRITICAL
BEFORE_EVENT	DMD	**Inspect and reinforce coastal drainage outfalls** Conduct structural inspections of Ballard Bunder Sluice, Mazgaon Outfall Gate, and Marine Drive Back Bay Sluices; install temporary sandbag barriers where clearance is limited.	A, B, C	HIGH
BEFORE_EVENT	ROADS	**Develop traffic diversion plans for arterial corridors** Map alternative routes for Marine Drive, Shahid Bhagat Singh Road, and Eastern Freeway; pre-position signage and traffic personnel to manage potential congestion during surge.	A, E	HIGH
BEFORE_EVENT	HEALTH	**Activate hospital surge capacity protocols** Validate emergency power, oxygen reserves, and isolation wards at the three hospitals in Ward A, the J.J. Hospital in Ward B, and the two facilities in Ward E; pre-assign additional medical staff for rapid deployment.	A, B, E	CRITICAL
DURING_MONITORING	SWM	**Monitor waste collection routes for blockage risk** Deploy field teams to inspect waste collection points along affected roads and ensure that debris does not impede drainage or pump intake structures.	A, B, C, D, E	MEDIUM
DURING_MONITORING	POLICE	**Establish hotspot crowd-control units** Station police units at identified chronic hotspots (e.g., Colaba Post Office junction, J.J. Hospital Junction, Byculla Police Station Junction) to manage pedestrian flow and assist evacuation if water levels rise.	A, B, E	CRITICAL
POST_EVENT	SWD	**Conduct rapid post-surge drainage assessment** Survey all pumping stations and drainage channels for sediment buildup; prioritize de-watering of the Byculla Lowland trough and other depressions identified in Ward E.	E	HIGH
POST_EVENT	HEALTH	**Implement post-event health surveillance** Activate disease surveillance units at the three Ward A hospitals and the two Ward E hospitals to monitor water-borne and vector-borne illness trends.	A, E	HIGH

2. AI GOVERNANCE & DATA PROVENANCE

AUDIT TRACE

AI Orchestration State: PARTIAL_DEGRADED (40402 ms)
Role 1 (Evidence Analyst): FAILED_AUTH (deterministic-rules-engine)
Role 2 (Impact Analyst): OPERATIONAL (openai/gpt-oss-120b)
Role 3 (Report Editor): FALLBACK_DETERMINISTIC (deterministic-rules-engine)

PREPARED BY:

Senior GIS & Hydrological Modeller
Disaster Management Support Group, MCGM

Signature & Municipal Seal

EXECUTIVE REVIEW & AUTHORISATION:

Director / Additional Municipal Commissioner
Disaster Management Department, MCGM Head Office

Authorisation Signature

STATUTORY DISCLAIMER & LIMITATIONS: This document constitutes a computerized multi-criteria hydrodynamic screening evaluation based on digital elevation modeling (Copernicus GLO-30 DEM) and BRIMSTOWAD storm drainage catchments. It is designed to assist administrative situational awareness and pre-positioning. It does NOT constitute an automated municipal proclamation, legal evacuation order, or public emergency dispatch. Official deployment of personnel, equipment, and public alarms requires formal sign-off by the Municipal Commissioner or designated Disaster Management Officer.